Example 9.13

Evaluate : $\int_0^{\frac{\pi}{2}} \left(\sqrt{\tan x} + \sqrt{\cot x}\right) dx$.

Solution

Let $I = \int_0^{\frac{\pi}{2}} \left(\sqrt{\tan x} + \sqrt{\cot x}\right) dx$. Then, we get

$$I = \int_0^{\frac{\pi}{2}} \left(\sqrt{\frac{\sin x}{\cos x}} + \sqrt{\frac{\cos x}{\sin x}}\right) dx = \int_0^{\frac{\pi}{2}} \frac{\sin x + \cos x}{\sqrt{\sin x \cos x}} dx = \sqrt{2} \int_0^{\frac{\pi}{2}} \frac{\sin x + \cos x}{\sqrt{2 \sin x \cos x}} dx$$

$$= \sqrt{2} \int_0^{\frac{\pi}{2}} \frac{(\sin x + \cos x) dx}{\sqrt{1 - (\sin x - \cos x)^2}} .$$

Put $u = \sin x - \cos x$. Then, $du = (\cos x + \sin x) dx$.

When $x = 0, u = -1$. When $x = \frac{\pi}{2}, u = 1$.

$$\therefore I = \sqrt{2} \int_{-1}^{1} \frac{du}{\sqrt{1 - u^2}} = \sqrt{2} [\sin^{-1} u]_{-1}^{1} = \sqrt{2} \left[\sin^{-1}(1) - \sin^{-1}(-1)\right] = \pi\sqrt{2} \quad .$$

Example 9.14

Evaluate : $\int_0^{1.5} \left[x^2\right] dx$, where $[x]$ is the greatest integer function.

Solution

We know that the greatest integer function $[x]$ is the largest integer less than or equal to x. In other words, it is defined by $[x] = n$, if $n \le x < (n+1)$, where n is an integer.

So, we get $$[x^2] = \begin{cases} 0 & \text{if} \quad 0 \le x < 1 \\ 1 & \text{if} \quad 1 \le x < \sqrt{2} \\ 2 & \text{if} \quad \sqrt{2} \le x \le 1.5 \end{cases}$$

We note that the above function is not continuous on $[0,1.5]$.

But, it is continuous in each of the sub-intervals $[0,1)$, $[1,\sqrt{2})$ and $[\sqrt{2},1.5]$; that is, it is piece-wise continuous on $[0,1.5]$. See Fig. 9.6. Hence, we get

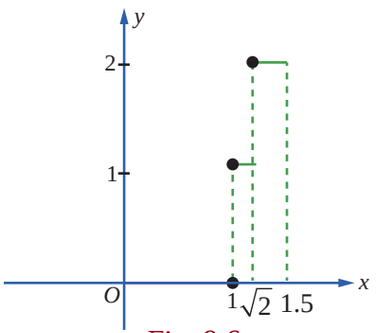


Fig. 9.6

$$\int_0^{1.5} \left[x^2\right] dx = \int_0^{1} \left[x^2\right] dx + \int_1^{\sqrt{2}} \left[x^2\right] dx + \int_{\sqrt{2}}^{1.5} \left[x^2\right] dx = \int_0^{1} 0 \, dx + \int_1^{\sqrt{2}} 1 \, dx + \int_{\sqrt{2}}^{1.5} 2 dx$$

$$= 0 + (x)_1^{\sqrt{2}} + (2x)_{\sqrt{2}}^{1.5} = \left(\sqrt{2} - 1\right) + \left(3 - 2\sqrt{2}\right) = 2 - \sqrt{2}.$$

Example 9.15

Evaluate : $\int_{-4}^{4} |x + 3| \, dx$.

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